Financial Advisor Agent in a Multi-Agent Financial Trading System

Vikrant Pandey Wee-Keong Ng Ee-Peng Lim
School of Computer Engineering
Nanyang Technological University SINGAPORE
awkng@ntu.edu.sg

Abstract

Agent-based technology has gained a lot of importance over recent years as is evident from the rapidly growing number of Websites using agents. In the future, the level of automation would be such that work delegated to an agent could be accomplished by collaborating with other agents. Conventional financial trading systems alert the user based on the analysis of financial instruments based on user inputs. In this work, we take it a step further; an agent may initiate actions up to a limit defined by the user on the belief that in light of the analysis information available, the user would have taken the same action. This paper describes a multi-agent financial trading system that helps to manage a financial portfolio by monitoring variables such as stock prices. Agents alert their owner or take autonomous actions when the information being monitored satisfies certain user-established criteria. All agents collaborate with one another to retrieve financial information, perform analysis and take appropriate actions such as investing at the best rates trying to optimize the client portfolio.

1. Introduction

Electronic commerce is a rapidly growing area accompanied by the growth of the Internet and the World Wide Web. As information on the Internet becomes more dynamic and heterogeneous, *software agents* have been touted as the new building blocks for a new Internet structure. We foresee a future whereby the Web is populated with as many software agents (involved in electronic commerce and other activities) as there are Web documents today. Organisations will not only establish Web presence via Web sites alone, they will also provide agents to interact with other agents. The level of automation would be such that work delegated to an agent could be accomplished by collaborating with other agents [4, 5].

Conventional financial trading systems have thus far been used only for the analysis of financial instruments and to suggest possible trades to the user. A user executes the trade using his user account in a financial trading institution. We take it a step further; a user representative agent could take actions up to a limit defined by the user on the belief that in light of the analysis information available, the user would have taken similar actions. This would be possible if various organisations provide agents to cooperate with other agents.

In this paper, we describe an expert Multi-Agent Financial Trading System (MAFTS) that demonstrates the usage of different agents provided by an organisation as applied to financial trading. In particular, we discuss the use of a *financial advisor agent* that helps a user manage a financial portfolio by monitoring variables such as *stock prices*. The agent is able to alert its owner or take autonomous action when the information being monitored satisfies certain user-established criteria. It collaborates with other agents to retrieve financial information, perform research on it and take appropriate actions such as investing at the best rates trying to optimise the client portfolio. Thus, this would pave the way for a level of automation where an entity gives the task to an agent and the agent executes the task on behalf of the entity. Focus shifts when developing highly specialised agents for doing a particular task leading towards perfection in that particular task.

The paper is organised as follows: Section 2 provides the background of the technology and financial tools used for the system. Section 3 discusses related work. Section 4 describes the architecture of the system, Section 5 discusses the implementation of the system and Section 6 concludes the paper highlighting the future work that could be done.

2. Background

The Common Object Request Broker Architecture (CORBA) has been chosen for implementation in order to speed up the development time and reduce development effort by benefiting from the *Naming, Trader, Location* and *Authentication* services it provides. It delivers *Enterprise*-ready solutions that are highly scalable, robust and secure.

0-7695-0680-1/00 $10.00 © 2000 IEEE

Concepts from *Business Technical Analysis* and *Moving Average Convergence Divergence* have been utilised in the design of the system. In this section, some insight and justification on these choices is provided.

CORBA as A Multi-agent Platform

CORBA is an open standard that allows many different types of hardware and software applications to communicate with one another no matter what language they are written in or where they are located on a network. It is a complete, distributed object computing framework that extends applications across networks, languages, component boundaries and operating systems.

CORBA is both an architecture and a specification for producing CORBA compliant Object Request Brokers or ORBs. A broker is analogus to the facilitator presented in architecture of agent interoperability [12]. Brokers mediate communication between objects or agents and provide services that facilitate their collaboration [1]. It is a meta-language where the role of the facilitator is played by the ORBs and the role of *agents* by the *objects* themselves. CORBA is an example of a pseudo-federated system in which assisted co-ordination takes place among the different agents. The advantage of using CORBA is that many of the object services are standardized and ready for the agents to use.

Business Technical Analysis

There are two basic types of market analysis techniques available to guide market players; *fundamental analysis*, which is based on forecasting the future price movements of a financial instrument based on economic, political, environmental and statistical factors and *technical analysis*, which is a method of predicting price movements and future market trends by studying charts of past market action [6]. Technical analysis is primarily used due to its quantitative nature. It is based on three underlying principles. The first principle states that *market action discounts everything*, which means that the actual price is a reflection of everything that is known to the market that could affect it such as supply and demand, political factors and market sentiment. The second principle is based on the assumption that *patterns exist* and technical analysis is used to identify patterns of market behavior, which have long been identified as significant. The third principle, *history repeats itself*, is based on the assumption that human psychology changes little over time thus leading to many repeated patterns.

Technical analysis provides market players with a variety of indicators to derive their trading strategies. The two basic type of indicators used are *confirmation/divergence indicators* and *momentum indicators or oscillators*. The *moving average convergence divergence* indicator, which is a type of a *momemtum indicator* is used in the MAFTS project.

Moving Average Convergence Divergence

The Moving Average Convergence Divergence (MACD) indicator is developed by Gerald Appel to trade 26 and 12-week cycles in the stock market [13]. MACD is a type of oscillator that can measure market momentum as well as follow or indicate the trend. It consists of two lines, the *MACD line* and the *signal line*. The *MACD line* measures the difference between a short exponential moving average and a long exponential moving average. The *signal line* is an exponential moving average of the MACD line. MACD oscillates above and below a zero line without upper and lower boundaries. The most common uses of MACD are to generate buy and sell signals, to indicate trend direction and to indicate the weakening of the upmove (Bullish Divergence) and weakening of the downmove (Bearish Divergence).

Buy and sell signals are generated when the MACD line and the signal line cross. A buy signal is generated when the MACD line crosses from below to above the signal line. The further below the zero line that this occurs the stronger the signal. Similarly, a sell signal is generated when the MACD line crosses from above to below the signal line. The further above the zero line this occurs, the stronger is the signal.

3. Related Work

Much work has been done in the field of software agents and financial engineering. The *Quo Vadis* system employs a multi-agent architecture and tries to integrate data-mining, knowledge discovery and soft-computing into a single framework [8]. The *expert* nature of MAFTS makes it different from the Quo Vadis system. Yuan [10] proposed a multi-agent approach as the key to solving dynamic Knowledge Discovery in Database (KDD) applications to reduce the manual effort required in these kind of applications through the demonstration of a cooperative information system for automating dynamic KDD applications from a large amount of stock data using multi-agent technology. A financial cybermarket with a bundle trading mechanism forms the discussion of [7]. A system is proposed where the order to buy/sell stocks occurs in bundles rather than a single stock in order to better balance the portfolio of the client. The Common Agent Request Broker Architecture or CARBA has been suggested as a dynamic architecture for multi-agent systems that is similar to CORBA [11]. It is based on the concept of the addition of knowledge and reasoning capabilities to CORBA, thus an attempt to develop a pure federated system with CORBA-like architecture. Chande [2] explains how technical analysis can be used to implement profitable trading systems. Rugiero [3] further develops on it providing suggestions on develop-

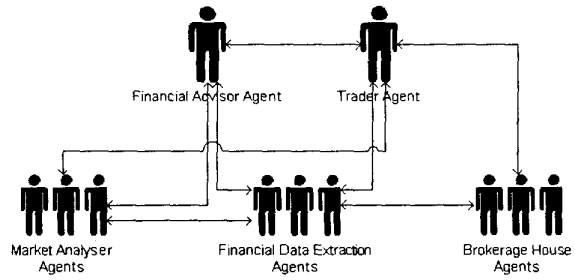

Figure 1. Multi-Agent Financial Trading System

ing state-of-the-art trading systems using neural networks, genetic algorithms and fuzzy logic. Lim and Wunsch [9] take a fuzzy perspective towards business technical analysis. A case study of trend prediction using moving averages has been discussed. Fuzzy logic has been suggested as the tool to specify precise thresholds for deciding on a particular course of action.

4. Architecture of the Multi-Agent Financial Trading System

We describe the architecture with the aid of a simple scenario on how trade takes place in the real world from the perspective of a common user.

A user first decides the amount to be invested and then reads through the newspapers and magazines to pick up a particular stock for investing. After that, the user monitors the stock daily untill the price and other market analysis factors match the user criteria. Then he buys/sells the stock on that day.

Highly specialised agents have been used in MAFTS to maximize returns in this information age. The agent-based financial trading system consists of Financial Advisor agents (FAAs), Market Analyser agents (MAAs), Financial Data Extraction agents (FDEAs), Brokerage House agents (BHAs) and Trader agents (TAs) as shown in Figure 1. FAA serves as the main link between the user and the system. It represents the user in the system. MAA does the research work; it analyses market trends for a particular equity to be monitored and generates real-time signals that can be used by the FAA for forming its trading strategies to gather and filter information from financial databases over the network. BHA is the representative of the Brokerage House. The FAA executes a trade with the BHA. Lastly, the TA functions like a yellow page; it helps different agents to locate one another.

A transaction starts with the FAA. A user, whom the agent represents, specifies the name of the stock to be monitored, the amount authorised and the investment policy. The FAA also asks the user to specify the type of MAA and BHA to be used. For instance the attributes describing the MAA could be *type, indicator, charges* and so on. Instances of type include *fundamental analyser, technical analyser*. The attributes for FDEA could be *type* (Free, Paid) and *charges*. Similarly, the attributes describing BHA could be *rating, commission* and so on.

The FAA then sends requests to the TA to get the references of agents satisfying the above criteria. The TA searches a list of registered agents and provides the reference of agents that match the criteria. Once the TA returns the appropriate MAA, the FAA passes the stock to be monitored to the MAA. The MAA contacts the FDEA providing it with the historical data as well as online data for that particular stock. It analyses the data and generates buy/sell signals required by the FAA. The FAA decides the trading strategy based on the signal provided by the MAA and contacts the BHA to execute the trading decisions. The next section describes these agents in more detail.

5. Implementation

Financial Advisor Agent

The Financial Advisor Agent (FAA) interacts with the user and co-ordinates with other agents on behalf of the user. The user gives authority to the FAA to invest in a particular share on the user's behalf and maximise the returns for the user. As an example, consider that the user gives the agent authority to invest *$1000* on Singapore Airlines shares. The amount to be invested is decided upon the policy choosen by the user, which could be:

- **Fixed Investment Policy:** The agent buys or sells shares worth a fixed amount whenever a trading signal is received. Under this policy the agent invests $100 each time the trading signal is received.
- **Proportional Investment Policy:** The agent buys or sells shares worth a proportional amount, proportional to the strength of the buying or selling signal generated. The signal generated is in the range $[-1, 1]$ with a positive value for a sell signal and a negative value for a buy signal. The user can specify the maximum limit that corresponds to the ceilings. In the example, if the user specifies the maximum limit to be $100 and if the trading signal generated is 0.6 then the agent sells holdings worth $60 dollars.

Market Analyser Agent

The Market Analyser Agent (MAA) analyses the market, generates buy/sell signals and passes them back to the Financial Advisor Agent. The MAAs are differentiated on the

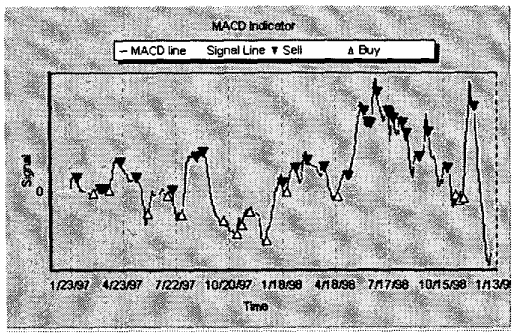

Figure 2. Technical Analysis Agent plot.

basis of the *type* (Fundamental Analyser, Technical Analyser, or combination). The *Technical Analyser* is further differentiated on the type of indicator used such as MACD, Stochastic Oscillator and so on. The MAAs use different techniques to analyse the market. The MACD indicator described in Section 2 is used as the MAA in the system. Figure 2 shows the plot that was obtained as a result of applying the MACD technique. The historical data of Singapore Airlines stock was taken from 1997 that consists of the closing prices of each day since then. The intersections between the MACD line and signal line were calculated and depending upon the rules as highlighted in Section 2, buy and sell intersection points were determined.

The intersection points from Figure 2 were then normalised in the range $[-1, 1]$ to produce the plot of Figure 3 indicating the buy/sell signals.

The generated signal is passed to the FAA in real-time. As soon as a buy/sell condition occurs, the FAA executes the trade, thus optimising the returns for the user by buying/selling following the trend.

Financial Data Extraction Agent

The Financial Data Extraction Agent (FDEA) extracts the information required by the MAA or the FAA. The data could be online information such as the trading taking place in major stock exchanges or it could be historical data pertaining to the particular equity. In the market, there are different sources of financial information; some are paid ones (e.g., Reuters, Bloomberg) while others are free (e.g., Yahoo, Infoseek). The difference is in the kind of information available. The paid sources keep information that is as old as 15 to 20 years. Besides, they also provide information on equities that are of a *restrictive access* status that is not available from the free sources. The other difference is that the information provided by the free sources is delayed by 15 to 20 minutes. In fact, most of the free sources receive their information from these paid sources. The FDEA developed for our MAFTS extracts financial information come from the Reuters server. In future, other agents can be developed that extract information from the free sources. In this section, we describe a Financial Data Extraction Agent that extracts historical and online stock information provided by the Reuters information server.

Reuters provides an Integrated Data Network Dynamic Data Exchange (IDNDDE) server for applications to extract financial information. The Reuters IDNDDE Server, like any other Dynamic Data Exchange (DDE) server, is used to transfer information between Windows applications. More specifically, IDNDDE is used to transfer information between the RT Data Manager and third party applications. The RT Data Manager is an interface to allow access to Reuters' global information and news databases. The Reuters FDEA acts as a DDE client and passes the required information to the other agents.The other agents use the data provided by the FDEA to do their analysis and accomplish their work.

Brokerage House Agent

A brokerage house is a company that acts as a broker, or intermediary, in the purchase/sale of investment securities. The Brokerage house has been implemented as a server, encapsulating customer accounts and their portfolio of securities. The Brokerage House Agent (BHA) allows the brokerage customers to place trade, check portfolio balances, compute market values, access quote feeds and so forth.

The Brokerage House comprises of four main entities: Asset, Holding, Account and the Brokerage House. Asset represents a particular equity with attributes such as *ticker* and *name*. An account consists of a number of holdings of a particular asset. The Brokerage House maintains a number of Accounts. The BrokerageHouse interface consists of two main methods *openAccount* and *loadAccount*. These methods allow other agents to open a new account or load an existing account. Trades are transacted against the account through asset buys and sells. An operation for finding out the market value of the account is also provided that does so by collaborating with the FDEA.

Trader Agent

The Trader Agent (TA) implements a simple *trader* interface that enables agents to register themselves with the trader. The registration information used by agents consists of a list of properties (name/value pairs). Agents can query the trader with a list of properties (name/value pairs) called criteria and obtain a list of *object references* that match the criteria set. Once an object reference is received, there is no need for the TA.

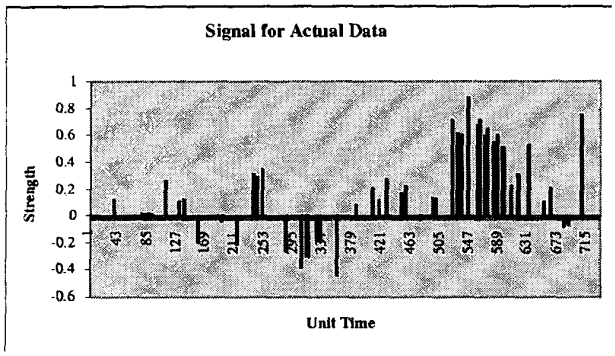

Figure 3. Signal generated from the Technical Analysis Agent plot.

6. Summary

The growing popularity of software agents and the fast emerging field of financial engineering played a vital role in the MAFTS project. The project aims to provide an infrastructure for a very large scale, distributed, agent based system with a focus on computational finance. The basic design of the system focuses on the application of agent technology to maximise the portfolio of a user. This paper describes how the task could be delegated to an agent and the agent in turn collaborates with other agents to achieve the goal of maximizing returns for the owner.

MAFTS explores the use of CORBA as the medium for implementing large-scale, robust and secure distributed systems that are inter-operable. The system also demonstrates the co-existence and co-operation of two different technologies for inter-application communication. It shows the communication between a Reuters Integrated Data Network Dynamic Data Exchange server and a CORBA server agent.

In future, neural networks, fuzzy logic and genetic algorithms based agents can be added to increase the degree of specialisation of these agents. The addition of new agents into the system is relatively much easier because of the framework provided by CORBA. The present system uses technical analysis based market analysis agents due to its quantative nature. It is more difficult to model fundamental analysis information due to its subjective nature. The modeling of fundamental analysis information can form a good topic of future research. Fundamental analysis based agents can be developed and a trading strategy incorporating both can be developed.

References

[1] M. KNAPIK, J. JOHNSON. *Developing Intelligent Agents for Distributed Systems: exploring architecture, technologies and applications,* McGraw-Hill Companies, Inc., 1998.

[2] TUSHAR S. CHANDE. *Beyond technical analysis: how to develop & implement a winning trading system,* John Wiley & Sons, Inc., 1997.

[3] MURRAY A. RUGIERO, JR. *Cybernetic Trading Strategies: Developing a Profitable Trading System with State-of-the-Art Technologies,* John Wiley & Sons, Inc., 1997.

[4] G. YAN, W. K. NG AND E. P. LIM. Toolkits for a Distributed, Agent-Based Web Commerce System, In *Proceedings of The International IFIP Working Conference on Trends in Distributed Systems for Electronic Commerce (TrEC'98),* Hamburg, Germany, July 1998.

[5] ABECOS HOMEPAGE. Centre for Advanced Information Systems, School of Applied Science, Nanyang Technological University, Singapore. *http://www.cais.ntu.edu.sg:8000/ abecos.*

[6] REUTERS FINANCIAL TRAINING SERIES. *An Introduction to Technical Analysis,* John Wiley & Sons (Asia) Pte Ltd, 1999.

[7] M. FAN, J. STALLAERT, AND A. B. WHINSTON. The Design and Development of a Financial Cybermarket with a Bundle Trading Mechanism,*In International Journal of Electronic Commerce Fall 1999, Vol. 4, No. 1*

[8] V. SRIKANTH. Intelligent Trading Systems: A Multi-Agent Hybrid Architecture In *IEEE/IAFE Conference on Computational Intelligence for Financial Engineering,* March 1999.

[9] M. H. LIM, D. WUNSCH. A Fuzzy perspective towards Technical Analysis - Case Study of Treand Prediction Using Moing Averages,*In IEEE/IAFE Conference on Computational Intelligence for Financial Engineering,* March 1999.

[10] S. T. YUAN. Multi-Agent Approach as a Catalyst to a Dynamic Financial Knowledge Discovery Process, *In Journal of Computational Intelligence in Finance),* July/August1999.

[11] W. JIAO, Z. SHI. A dynamic architecture for multi-agent systems, *In Technology of Object-Oriented Languages and Systems,* 1999.

[12] M. R. GENESERETH. Interoperability: An Agent-Based Framework. In *Artificial Intelligence Expert, pp. 34-40,* March 1995.

[13] T. HARTLE. Moving Average Convergence/Divergence, *In Technical Analysis of Stocks and Commodities Volume 9 pp. 104,*1991